Nuclear Physics

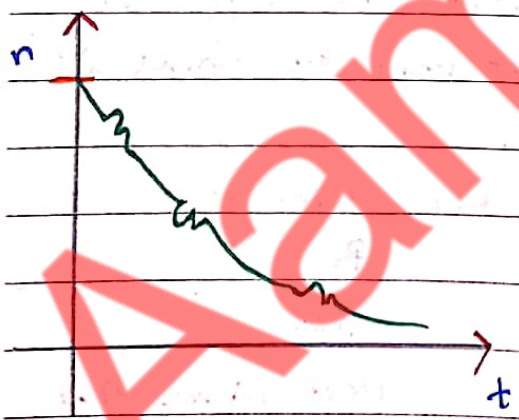
Radioactive Decay

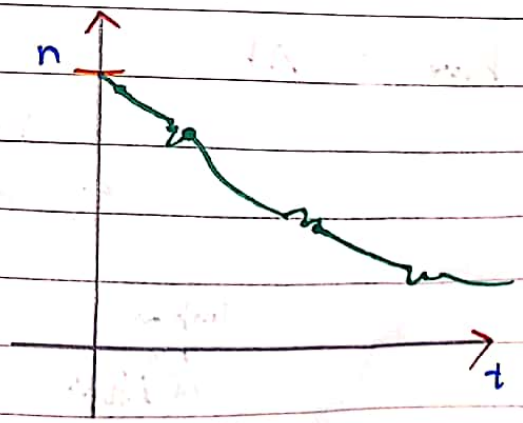
A process in which an unstable element emits radiation to become stable is called "Radioactive Decay". The element which emits radiations is called "Radioactive Element" & emitted radiations are called "Radioactive Radiations"

it is a
i- Random Process
ii- Spontaneous Process

→ There are few unstable elements.

→ Radioactive element, it's atoms decay without any sequence or pattern so is called a "Random Process" e.g if 40 atoms in a radioactive element & it's half life is 5s so after 5s 20 atom left but this does not mean equal decay per second this is called a "Random Process"





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→ Rate of Decay of Radioactive atoms does not remain constant

(OR)

No of atoms decayed per second does not remain constant

↳ Statement of Random Process

→ Physical Conditions does not effect the rate of decay thus we call it as a "Spontaneous Process"

→ Rate of Decay is not effected by change in physical conditions such as temperature & pressure

↳ Statement of Spontaneous Process

Explanation of Graphs

→ In 1st graph, we have drawn rate of decay graph for radioactive element (no of atom vs time graph)

→ For 2nd graph we have changed the physical conditions

↳ Both these graph tell that radioactive decay is a random & spontaneous ~~graph~~ process as the shape of graph (generally same shape) is same (so spontaneous) and the spikes tell us that it is a random process (different decay at different time intervals).

Radioactive Element

Initial no of atoms = N at t = 0

No of atoms decayed = ΔN

Decaying time = Δt

$\Delta N \propto -N$ — i

$\Delta N \propto \Delta t$ — ii

i ⇒ The more the initial no of atoms the more the atoms will be decayed.

e.g) Same radioactive elements.

	A	B
Initial No of Atoms	500	5000 ✓

More atoms that are decayed

→ The - element

ii ⇒ The decay

→ There

Derivation not in syllabus

Activity

Total No

→ Unit

→ By decaye random

Decay

decay

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→ The -ve sign shows that the no of atoms of radioactive elements decreases.

ii ⇒ The more the decaying time the more the atoms will be decayed

→ These two expressions are general for each radioactive elements

$$\Delta N \propto -N(\Delta t)$$

$$\frac{\Delta N}{\Delta t} \propto -N$$

$$\frac{\Delta N}{\Delta t} = -(\text{constant})N$$

Decay Constant (λ)

$$\frac{\Delta N}{\Delta t} = \text{Activity} = A$$

* tells us how many atoms decayed in one second

Derivation not in syllabus

$$\boxed{A = -\lambda N}$$

N = Initial No of Atoms

Activity

Rate of Decay of radioactive atoms is called "Activity"

(OR)

Total No of atoms decayed in unit time is called "Activity"

→ Unit of Activity is "s^{-1}" (Becquerel / Bq) — special Name for Activity

→ By Activity we can also predict the no of atoms decayed in the next second, it also changes randomly so random process

Decay Constant (λ)

Probability of each atom of radioactive element to decay in unit time is called "Decay Constant of Radioactive Element"

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→ Decay constant is a fixed value for all radioactive element and different from others

→ It is a property of radioactive element

→ By decay constant, we get the chance of each atom to decay in the next second

→ It's unit is s^{-1} (No special Name)

e.g. Radioactive element => $\lambda = 0.2 s^{-1}$ it means the chance of each atom to decay is 20%.

N = 1000 atoms

=> $\lambda \neq 1$ as not all atoms can decay.

Always less than 1

$A = -\lambda N$

$= -0.2 \times 1000$

$= -200$ Bq — to decay in the next second

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Nuclear Physics

Radioactive Decay

$$\boxed{A = -\lambda N}$$

$$\boxed{N = N_0 e^{-\lambda t}}$$

N_0 = Initial No of Atom
λ = Decay Constant
N = No of atoms left
t = Time taken for decay
e = Exponential constant

$$A = -\lambda N$$

$$\frac{\Delta N}{\Delta t} = -\lambda N$$

$$\frac{\Delta N}{N} = -\lambda (\Delta t)$$

$$\int \frac{\Delta N}{N} = \int -\lambda (\Delta t)$$

$$\int_{N_0}^{N} \frac{\Delta N}{N} = -\lambda \int_{t=0}^{t=t} \Delta t$$

$$\left| \ln N \right|_{N_0}^{N} \quad -\lambda \left| t \right|_{0}^{t}$$

$$\ln N - \ln N_0 = -\lambda (t - 0)$$

$$\ln \left(\frac{N}{N_0} \right) = -\lambda t$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\boxed{N = N_0 e^{-\lambda t}}$$

$$\frac{N}{t} = \frac{N_0}{t} e^{-\lambda t}$$

$$\boxed{A = A_0 e^{-\lambda t}}$$

$$\boxed{A = A_0 e^{-\lambda t}}$$

Q. $N_0 = 1000$ atoms
$\lambda = 0.2\,s^{-1}$
$t = 5\,s$
$N = ?$

$$N = N_0 e^{-\lambda t}$$

$\rightarrow$ 367.8 atoms

$$N = N_0 e^{-\lambda t}$$

$$\Rightarrow 1000\, e^{-(0.2 \times 5)}$$

$$= 367 \text{ atoms}$$

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Half Life $\left(\frac{T_1}{2}\right)$

Time in which half of atoms, of a radioactive element decay is called "Half Life" of the decay.

e.g) 8000 atoms in 5 min, 4000 atoms in 5 min, 2000 in 5 min .....

→ It is represented by $\left(\frac{T_1}{2}\right)$

$$\boxed{T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}}$$

$$\boxed{N} = N_0 e^{-\lambda \boxed{t}}$$

$t = T_{\frac{1}{2}}$ $\quad N = \frac{N_0}{2}$

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{\frac{1}{2}}}$$

$$\frac{1}{2} = e^{-\lambda t_{\frac{1}{2}}}$$

$$\ln\left(\frac{1}{2}\right) = -\lambda T_{\frac{1}{2}}$$

$$-\lambda T_{\frac{1}{2}} = \ln(2)^{-1}$$

$$\cancel{-}\ln(2) = \cancel{-}\lambda T_{\frac{1}{2}}$$

$$\boxed{T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}}$$

Derivation is in syllabus

Radioactive Element

$t = 0$ $\quad N = N_0$

$t = T_{\frac{1}{2}}$ $\quad N = \frac{N_0}{2} = \frac{N_0}{(2)^1}$

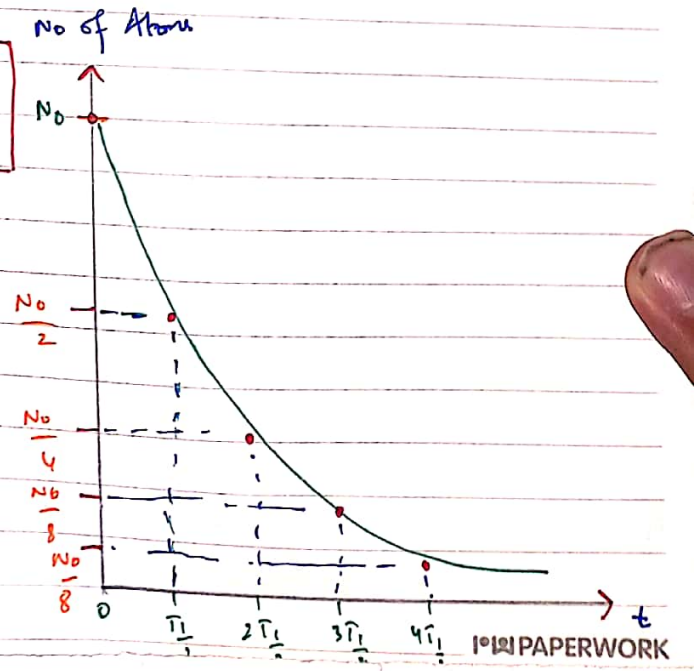
$t = 2\left(\frac{T_1}{2}\right)$ $\quad N = \frac{N_0}{4}$ $\quad$ Next half life $= \frac{N_0}{(2)^2}$

$t = 3\left(\frac{T_1}{2}\right)$ $\quad N = \frac{N_0}{8}$ $\quad$ 3rd Half life $\frac{N_0}{(2)^3}$

So general expression,

$$\boxed{t = n\left(\frac{T_1}{2}\right)} \qquad \boxed{N = \frac{N_0}{(2)^n}}$$

→ When atoms are decayed, new atoms are also formed



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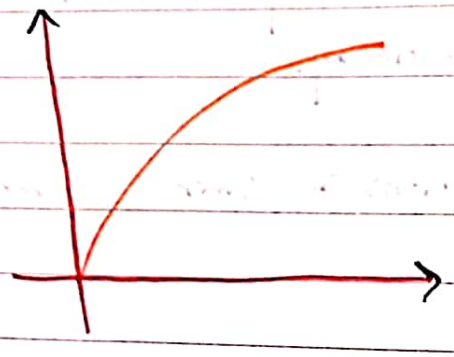
→ If asked to tell / draw the graph for newly formed ~~graph~~ atoms



⇒ opposite to decaying graph

Q.1

$A = 4.02 \times 10^{13}$ Bq

$N = 3.84 \times 10^{24}$ atoms

a) i- $\lambda = ?$

ii- $T_{\frac{1}{2}} = ?$

Q.2

Stronium - 90

$T_{\frac{1}{2}} = 28$ years

$A = 6.02 \times 10^{9}$ Bq

(Density) $\rho = 2.54\ g\,cm^{-3}$

a) $\lambda = ?$

b) $m = ?$

c) $V = ?$

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Atomic Mass Unit (u)

A unit used for the smaller particles

$u = \frac{1}{12}$ (mass of C-12 atom)

$u = 1.66 \times 10^{-27}$ kg

→ Amu is also an unit of mass in nuclear physics we use Amu for the smaller particles

→ We represent Atomic Mass Unit by "u"

e.g

$^{2}_{1}H$ $^{97}_{40}Zr$

Mass of Zuconium = 97.0980 u
Mass of Deutrium = 2.01410 u
Mass of Electron = 0.00055 u
Mass of Proton = 1.00728 u
Mass of Neutron = 1.00867 u

} Different Masses

↳ If we put the value of u then the answer in kg will be same for both (thats why same mass) but kg is a bigger unit which is not that accurate for smaller particles.

→ There is very small difference in masses of proton & neutron (not same)

→ A deutrium atom consists of 1 e^-, 1 proton & 1 neutron

↳ Total Mass = 2.01650 u (0.00055 + 1.00728 + 1.00867)

↓

Outside the deutrium atom, mass of constituent particles

→ However there is a difference in mass calculated & measured of the deutrium atom, this is called "Mass Defect"

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Mass Defect (Δm)

Difference in total mass of constituent particles and mass of an atom is called "Mass Defect of the Atom"

→ It is for all the atoms and for all atoms there will be a difference in mass of constituent particles of an atom

$$\boxed{\Delta m = (m_p + m_n + m_e) - m_A}$$

→ Mass of particle x No of particle ⟹ then add the mass of particles to get the total mass of particles.

For deutrium,

$$\Delta m = 2.01650 - 2.01410$$
$$= 0.00240\,u$$

For Zirconium,

$p = 40$
$n = 57$
$e = 40$

$$\Delta m = (1.00728 \times 40) + (0.00055 \times 40) + (1.00867 \times 57) - 97.0980$$
$$= 97.8074 - 97.0980$$
$$= 0.70944$$

→ Every atom has mass defect due to the binding energy

Binding Energy

$$E = (\Delta m)c^2$$

Δm = Mass defect
c = Speed of light

Energy required to split an atom into it's constituent particles is called it's "Binding Energy"

→ According to this expression energy can be converted into mass and vice versa

→ To break the atom we require energy which is the binding energy

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(small particles)

→ The energy provided will be converted into mass this is why the value of mass will increase and create the mass defect

→ When we are finding the binding energy we will convert the mass defect into kg and then use the formulae.

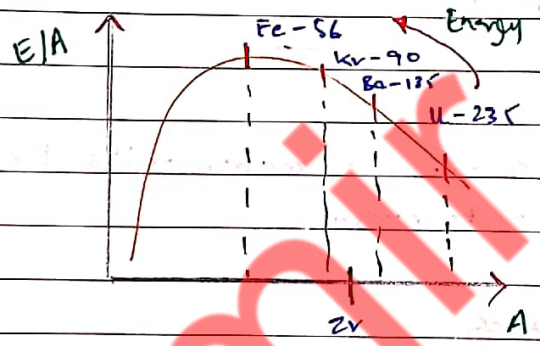
→ If asked to prove that 1 amu = ... energy then use the formula of binding energy with Δm as the mass of atom

Binding Energy per Nucleons

$$\text{Binding Energy per Nucleons} = \frac{E}{A}$$

E = Binding Energy
A = Nucleon No

→ Binding energy per nucleon represents the stability of atom, the more the energy, the more stable the atom will be



→ If asked to label Fe = 56 the most stable atom

→ Axis will not be labelled so (scale) after Fe is asked to label Zr, label it further from Fe as Fe is the reference

→ Due to Nuclear fission the stability increases

→ For unstable atom it needs to release energy to become stable.

Nuclear Fission

A process in which a heavy nucleus breaks into 2 light nuclei with release of energy is called "Nuclear Fission"

$$^{235}_{92}U + ^{1}_{0}n \longrightarrow ^{144}_{56}Ba + ^{90}_{36}Kr + 2^{1}_{0}n + \text{Energy}$$

→ Examiner will then ask to label Ba, Kr & U

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		E/A Binding energy per nucleon
$^{235}_{92}U$		1.2768×10^{-12} J
$^{144}_{56}Ba$		1.3621×10^{-12} J
$^{90}_{36}Kr$		1.3854×10^{-12} J

→ If asked to find the released energy

Method

1. Multiply the E/A x the nucleon No of Atom
2. Add Energies of Ba + Kr and the substract it from the energy of U

→ Then if asked to find mass use $E = (\Delta m)c^2$

Nuclear Fusion

A process in which two light nuclei combine to form a heavy nucleus with release of energy is called "Nuclear Fusion"

$$\boxed{^{2}_{1}H + ^{2}_{1}H \longrightarrow ^{3}_{2}He + ^{1}_{0}n + \text{Energy}}$$

→ We donot consider the Binding energy of neutron as it's E/A cz it's released and is not part of the nucleus

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